Freight Rate Prediction

Name: Hariprasanth Paranjothy | **Model:** Blended gradient-boosted trees on rate-per-mile | **Holdout MAE:** $111.53 (4.87% MAPE)

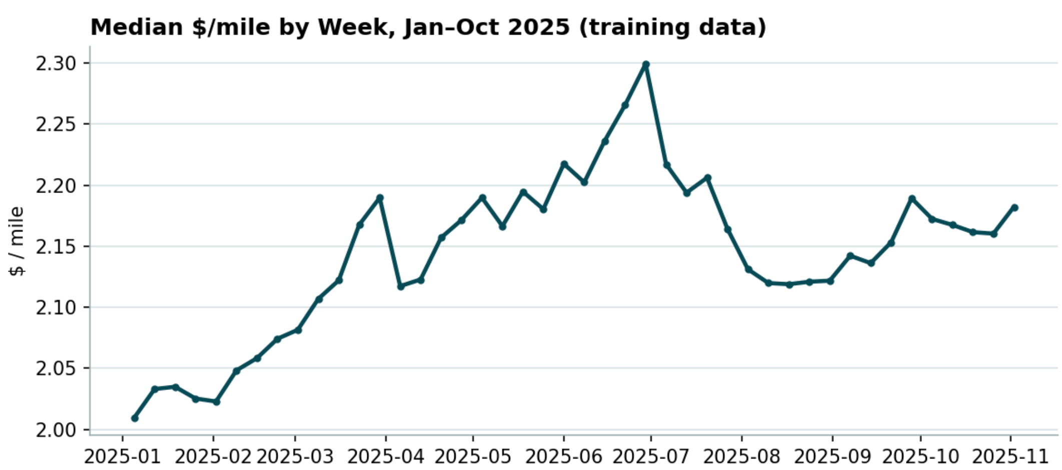
1. Problem & Data

The task is to predict the posted freight rate ($) for a load from its route, equipment type, shipment weight, calendar date, and two market signals (market_index, quote_signal).

File	Rows	Date range	Role
train-test.csv	48,000	Jan 1 to Oct 31, 2025	Labeled development data
validation.csv	12,000	Nov 1 to Dec 31, 2025	Unlabeled; scored by Spotter
december-chart-inputs.csv	31	Dec 1 to 31, 2025	Fixed Lexington→Fort Wayne lane, chart only

2. Exploratory Data Analysis

- Distance dominates price: corr(distance, posted_rate) ≈ 0.91.
- Rate-per-mile ($/mile) is a far more stable target than raw dollars. Median is about $2.15/mi, and equipment ordering runs Reefer > Flatbed > Dry Van.
- market_index behaves as a market-wide daily signal: within-day standard deviation (≈0.025) is far smaller than its overall standard deviation (≈0.17).
- quote_signal is closer to load-level noise: within-day std is nearly as large as overall std, so it's better modeled with lane/equipment priors than date.
- $/mile drifts with the seasons: a summer peak (Jun to Jul), a late-summer trough, and a rise back into Q4 (chart below). Since Nov to Dec sit past the training window, the model has to extrapolate this trend, not just interpolate it.



- validation.csv contains pickup/delivery cities absent from train-test.csv, so categorical encodings needed a sane fallback (a smoothed global/segment prior) instead of crashing or defaulting to zero.

3. Data-Quality Issues & Fixes

Issue	Evidence	Fix
~292 negative weight values (0.6%)	Magnitude distribution matches valid weights, which looks like a sign-flip error	abs(weight); flag retained as a feature (weight_was_negative)
~300 to 370 missing weight / market_index values	Scattered, not concentrated on specific dates or lanes	Weight: equipment-median. Market index: daily median → monthly median → global median cascade
A handful of short lanes list distance far above straight-line distance (e.g., Lubbock↔Austin)	distance ÷ haversine ratio up to ≈9.6× vs. a typical ≈1.2×	Left as-is. The same (possibly inflated) distance is given consistently at both train and predict time, which is consistent with a practical minimum-mileage billing floor, so it's documented rather than "corrected."
December chart input has no lat/lon, market_index, or quote_signal at all	This is by design, since these represent live market state that a future load can't have yet	lat/lon backfilled from a city-coordinate table; market_index pulled from validation.csv's Nov to Dec daily medians; quote_signal pulled from this lane's historical median

4. Feature Engineering

Temporal: month, day-of-week, day-of-year, weekend / month-start / month-end flags, sin⁄cos cyclical encodings.

Geographic: Δlat⁄Δlon, haversine distance, bearing, distance⁄haversine ratio, log⁄sqrt distance.

Interactions: distance×market_index, distance×quote_signal, market_index×quote_signal, and an expected_base_rate = distance×quote×market composite.

Target encodings: lane / pickup / delivery / equipment rate-per-mile priors, fit on the training fold only and Bayesian-smoothed toward the global rate so sparse lanes don't overfit.

Missingness flags: weight_missing, market_index_missing, quote_signal_missing kept as their own binary features, so the model can learn to trust imputed values a little less.

5. Train / Validation Split

Chronological, not random: train on 2025-01-01 to 2025-08-31, holding out 2025-09-01 to 2025-10-31.

A random K-fold split would let the model see both sides of the same week and overstate accuracy, since the real target (Nov to Dec) is entirely out-of-time relative to training. A trailing two-month holdout mimics the actual forecasting gap the submission faces. After holdout evaluation, all models are retrained on the full Jan to Oct dataset before generating the submitted predictions.

6. Model Selection: What Was Tried

Reframing the target as $/mile (then × distance to recover dollars) stabilizes the target across short vs. long hauls and improved every model tried. Two rounds of experiments on the Sep to Oct holdout:

Round 1: Establishing a Baseline

Model	MAE ($)
LGBM predicting $/mile × distance	148.3
CatBoost	138.8

Model	MAE ($)
Random Forest	138.5
Extra Trees	134.1
Extra Trees + RF + CatBoost (equal blend)	127.3
Best: 50% LGBM($/mi)×dist + 50% CatBoost	123.4

Round 2: Tuning Objectives and Blend Weights

Model	MAE ($)
CatBoost on $/mile × distance	115.4
LightGBM on $/mile × distance (MAE objective)	113.6
50% LGBM(rpm×mi) + 30% CatBoost(rpm×mi) + 20% Extra Trees(rpm×mi)	111.5

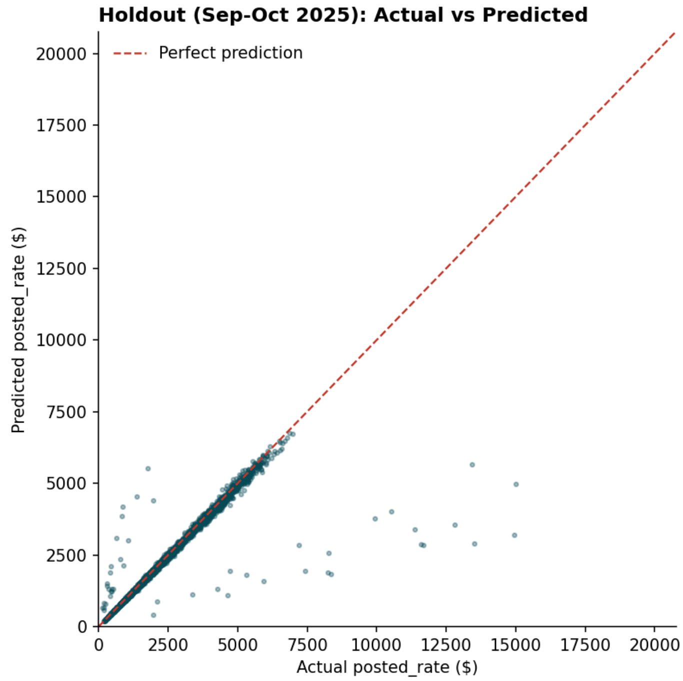
Switching both boosters' training objective to MAE and re-weighting the blend toward LightGBM took MAE from $123.4 to $111.5, roughly a 25% improvement over the very first single-model baseline.

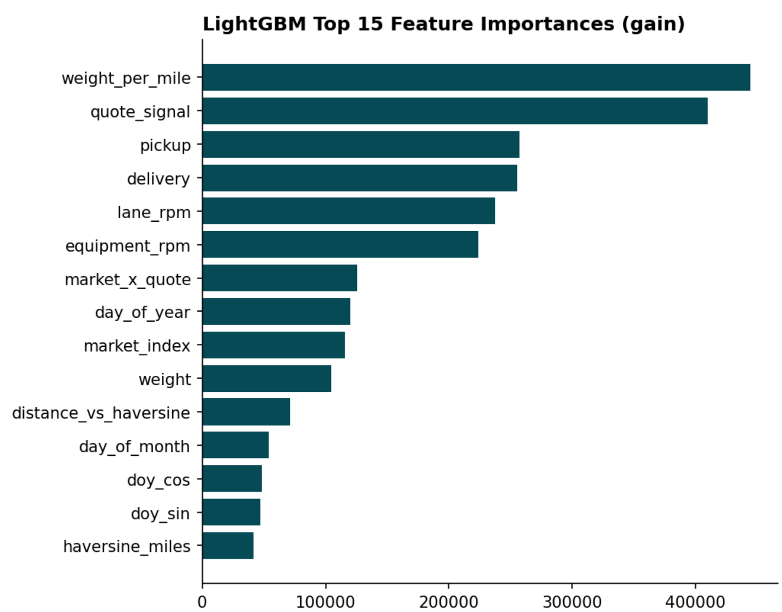
Final blend: 50% LightGBM ($/mi, MAE objective) + 30% CatBoost ($/mi, MAE objective) + 20% Extra Trees ($/mi).

Why trees, and why a blend: freight pricing is non-linear (equipment premiums, lane effects, and seasonal×market interactions don't fit a linear model well) and lane cardinality is high, which gradient-boosted trees handle natively as categoricals. Blending two differently-regularized boosters with a bagged Extra Trees model reduces variance without hand-tuning any single model further.

7. Final Model Performance

Metric	Value
MAE	$111.53
RMSE	$634.79
MAPE	4.87%
R^2	0.827





8. Where the Model Struggles

The headline MAE understates how tight the typical prediction is, and overstates how the model handles rare loads:

- Median holdout error is just $39; the 90th percentile is $137.
- But about 1.2% of holdout loads carry errors above $1,000, concentrated in loads whose actual rate-per-mile (median ≈ $5.79/mi) runs roughly 2.7× the typical $2.16/mi.
- These read like premium, expedited, or spot-surge loads that aren't well explained by the available features (route, weight, calendar, two market signals). The model regresses toward the lane's normal price rather than chasing rare spikes. That's the right behavior for minimizing MAE, but it means unusual premium loads get systematically under-priced.

9. December 2025 Forecast: Methodology

The Lexington → Fort Wayne / 360 mi / Dry Van / 32,000 lb lane ships with no lat/lon, market_index, or quote_signal at all, since those represent live market state a future load can't have yet.

- lat/lon: backfilled from the city-coordinate table built during training.
- market_index: pulled from the daily median of validation.csv for the matching December dates. validation.csv already spans Nov to Dec across many other lanes, and the EDA showed market_index is a market-wide daily signal, so this is a principled proxy, not a guess.
- quote_signal: pulled from this specific lane's historical median in the training data, since quote_signal proved to be lane/load-level rather than date-level.
- The same blended rate-per-mile model then runs on these enriched rows and multiplies by the fixed 360 miles.